

Question Bank

Course: Generative AI (25ECAC307)

Chapter 1: Introduction to NLP

L2 Level Questions (Understanding)

S.No	L2 - Level Questions
1	Illustrate with an example the different types of tokenization.
2	Explain why stemming and lemmatization are both important for text processing in NLP. How do they differ in their approach?
3	Imagine you're building a sentiment analysis system for social media posts. Describe the steps you would take in preprocessing the data, including tokenization, stop word removal, and either stemming or lemmatization.
4	Describe the difference between the Continuous Bag of Words (CBOW) and Skip-Gram models in Word2Vec.
5	Explain how a Dependency Tree represents sentence structure. What does each node and edge signify?
6	Why is POS tagging important in Natural Language Processing? What role does it play in understanding the syntactic structure of a sentence?
7	Define Natural Language Processing and list at least four real-world applications.
8	What is the difference between a corpus and a vocabulary in NLP text preprocessing?
9	Explain the concept of Bag of Words (BoW) representation with a simple example.
10	Explain the difference between GloVe and FastText word embedding techniques.

L3 Level Questions (Applying / Analyzing)

S.No	L3 - Level Questions
1	How can the given text document be transformed into vectors using One-Hot Encoding, Bag of Words, and TF-IDF?
2	Considering the sentence "iNeuron Company Is Related to Data Science", explain the working of CBOW and Skip-Gram models in Word2Vec.
3	Given a sentence, how would you use a Dependency Tree to identify the grammatical structure and relationships between words? Walk through the process of creating a Dependency Tree manually.
4	Imagine you're designing a chatbot that needs to understand locations and dates mentioned by users. How would you use both POS tagging and NER to improve its understanding of user input?
5	Compare and contrast Word2Vec, GloVe, and FastText embeddings in terms of how they handle out-of-vocabulary words.
6	Given the sentence "The quick brown fox jumps", perform POS tagging and identify each word's part of speech.
7	Analyze how TF-IDF overcomes the limitations of a simple word-count-based Bag of Words representation.
8	Design a text preprocessing pipeline for a spam-detection system, justifying each step (tokenization, stop-word removal, stemming/lemmatization).
9	How would Named Entity Recognition (NER) be applied to extract structured information from unstructured news articles? Illustrate with an example.
10	Evaluate the trade-offs between stemming and lemmatization when preprocessing text for a search engine application.

Chapter 2: Sequence-to-Sequence and Transformer Models

L2 Level Questions (Understanding)

S.No	L2 - Level Questions
1	What is Machine Translation (MT)? What are the primary steps of MT in Natural Language Processing (NLP)?
2	How does an LSTM store long-term dependencies in a sequence?
3	How does the attention mechanism address the limitations of Seq2Seq models, particularly the fixed-length context vector?
4	What is a variational autoencoder (VAE), and how does its architecture differ from a standard autoencoder?
5	Explain the difference between Rule-Based, Statistical, and Neural Machine Translation approaches.
6	Describe the basic architecture of an encoder-decoder (Seq2Seq) model.
7	What is the role of the context vector in a Seq2Seq model?
8	Explain the working principle and architecture of the attention mechanism.
9	What is an autoencoder? Describe its basic architecture and components.
10	Explain the architecture and purpose of a decoder in sequence generation tasks.

L3 Level Questions (Applying / Analyzing)

S.No	L3 - Level Questions
1	Is it possible to use Artificial Neural Networks (ANN) to process and analyze sequential data in Natural Language Processing (NLP)? Illustrate with the help of an example.
2	How does forward and backward propagation work in a Recurrent Neural Network (RNN) for the given dataset? Provide a detailed explanation with necessary equations.
3	What are the key components of a Seq2Seq model? How does a Seq2Seq model differ from a standard RNN?
4	Analyze the given figure and explain the concept of sequence-to-sequence (Seq2Seq) modeling.
5	You are tasked with improving the performance of a Seq2Seq machine translation model that uses a fixed-length context vector. How would you integrate an attention mechanism to enhance the translation accuracy? Walk through the steps of adding attention to the model.
6	How does an autoencoder learn to reconstruct data?
7	Compare Rule-Based Machine Translation, Statistical Machine Translation, and Neural Machine Translation in terms of accuracy, scalability, and data requirements.
8	Analyze how a Variational Autoencoder differs from a standard autoencoder in its latent space representation and training objective.
9	You are designing a text summarization system using a Seq2Seq model. Explain how you would incorporate attention to handle long input documents.
10	Evaluate the impact of vanishing gradients on RNN-based Seq2Seq models and how attention mechanisms help mitigate this issue.

Chapter 3: Generative Adversarial Networks

L2 Level Questions (Understanding)

S.No	L2 - Level Questions
1	Explain the architecture of a Generative Model. What are the key differences between generative and discriminative models?
2	How does a generative model learn the data distribution? Why are discriminative models more efficient for classification tasks?
3	What is a discriminative model? Which loss functions are commonly used in discriminative models?
4	Define a Generative Adversarial Network (GAN) and explain the role of the generator and discriminator.
5	Explain the concept of adversarial training in GANs.
6	List and briefly describe three types of GAN architectures.
7	What is the difference between Pure Statistical Language Models and Neural Language Models?
8	Explain the loss function used in a standard GAN.
9	Describe the concept of mode collapse in GAN training.
10	What are conditional GANs (cGANs), and how do they differ from vanilla GANs?

L3 Level Questions (Applying / Analyzing)

S.No	L3 - Level Questions
1	What type of GAN would you use for realistic image generation, and why?
2	Differentiate between Discriminative and Generative models with an example.
3	Can a model be both generative and discriminative? Give examples. What type of GAN would you use for converting satellite images to street maps?
4	Analyze how the minimax game between generator and discriminator drives GAN training toward equilibrium.
5	Compare DCGAN, StyleGAN, and CycleGAN in terms of architecture and typical use cases.
6	Design a GAN-based approach to generate synthetic medical images for training a diagnostic classifier, and justify your choice of GAN type.
7	Evaluate the challenges faced when training GANs, such as mode collapse and training instability, and suggest possible remedies.
8	How would you apply a CycleGAN to convert satellite images into street maps? Explain the underlying architecture.
9	Analyze the difference between Pure Statistical and Neural Language Models in terms of handling long-range dependencies.
10	Given a text-generation task with limited labeled data, justify whether a generative or discriminative model would be more suitable.

Chapter 4: Large Language Models

L2 Level Questions (Understanding)

S.No	L2 - Level Questions
1	What is a Large Language Model (LLM), and how does it function? Discuss its key characteristics and illustrate with examples some real-world applications across different domains.
2	Explain the concept of prompting in Large Language Models.
3	How do different prompting techniques (zero-shot, few-shot, and chain-of-thought prompting) affect the model's output and performance?
4	What are adapters in the context of Large Language Models?
5	How does LoRA reduce computational costs while maintaining model performance?
6	How does LoRA balance parameter efficiency and model performance, and what are its advantages and potential limitations compared to traditional full fine-tuning?
7	Explain the transformer-based architecture that underlies most modern LLMs.
8	What is the difference between pretraining and fine-tuning in the context of Large Language Models?
9	Describe the concept of tokenization used by LLMs such as GPT (e.g., byte-pair encoding).
10	Explain the concept of an adapter module in the context of parameter-efficient fine-tuning.

L3 Level Questions (Applying / Analyzing)

S.No	L3 - Level Questions
1	Describe the Low-Rank Adaptation (LoRA) technique for fine-tuning Large Language Models.
2	Differentiate between BERT and GPT in terms of architecture, training objective, and typical applications.
3	How do zero-shot, few-shot, and chain-of-thought prompting differ in their approach to eliciting desired outputs from an LLM? Illustrate with examples.
4	Design a prompting strategy for a customer-support chatbot to improve response accuracy on domain-specific queries.
5	Analyze how LoRA reduces the number of trainable parameters compared to full fine-tuning, using matrix decomposition concepts.
6	Evaluate the trade-offs between using full fine-tuning versus LoRA-based adaptation for deploying an LLM on a resource-constrained device.
7	Compare BERT's masked language modeling objective with GPT's autoregressive objective, and explain how each affects downstream task suitability.
8	How would you select between BERT and GPT for a question-answering system? Justify your answer with reasoning about their architectures.
9	Analyze the effect of chain-of-thought prompting on the reasoning capability of an LLM for a multi-step arithmetic problem.
10	Design an adapter-based fine-tuning pipeline for adapting a pretrained LLM to a legal-document summarization task.

Chapter 5: Diffusion Models

L2 Level Questions (Understanding)

S.No	L2 - Level Questions
1	Why are Transformer models generally preferred over RNNs and LSTMs for natural language processing tasks?
2	What are the key features of Transformer models that make them effective for NLP tasks?
3	Analyse the working of the Transformer encoder-decoder architecture with the help of a concrete example, highlighting how information flows from input to output through self-attention, cross-attention, and feed-forward layers.
4	Explain the architecture and training strategy of BERT (Bidirectional Encoder Representations from Transformers). How does its bidirectional attention mechanism differ from traditional left-to-right models, and what advantages does this offer for downstream NLP tasks?
5	Explain the architecture and training methodology of Generative Pretrained Transformers (GPT). How does its autoregressive nature influence its performance in text generation tasks?
6	Compare and contrast BERT and GPT in terms of their architectures, training objectives, and applications.
7	Analyze the working principles of diffusion models in generative modeling. How does the iterative denoising process contribute to high-quality sample generation, and how do diffusion models compare to GANs in terms of training stability and output diversity?
8	Explain the basic working principle of a diffusion model, including the forward (noising) and reverse (denoising) processes.
9	What is the difference between continuous and discrete diffusion models?
10	Describe the difference between deterministic and stochastic diffusion processes.

L3 Level Questions (Applying / Analyzing)

S.No	L3 - Level Questions
1	How does Multi-Head Attention enhance the model's ability to capture different aspects of input relationships? Illustrate your answer with an example.
2	Choose the appropriate type of diffusion model (continuous or discrete) for a text-generation task, and justify your choice.
3	Differentiate between deterministic and stochastic diffusion models in terms of sample diversity and reproducibility.
4	Differentiate between continuous and discrete diffusion models with respect to their application in image versus text generation.
5	Differentiate between linear and non-linear noise schedules in diffusion models, and analyze their effect on generation quality.
6	Analyze how the iterative denoising process in diffusion models contributes to high-quality sample generation compared to single-step generative models.
7	Compare diffusion models and GANs in terms of training stability, output diversity, and computational cost.
8	Design a diffusion-based pipeline for generating synthetic handwriting samples, explaining the choice of noise schedule.
9	Evaluate the impact of the number of diffusion timesteps on the trade-off between generation quality and inference speed.
10	Given a scenario requiring both high sample diversity and stable training, would you choose a diffusion model or a GAN? Justify your answer.

Chapter 6: Vision Language Models and Multimodal Generative AI

L2 Level Questions (Understanding)

S.No	L2 - Level Questions
1	What is Multimodal AI? Explain the importance of integrating vision and language in AI systems.
2	Describe joint vision-language representation learning. How does it help improve multimodal understanding?
3	Explain the architecture of CLIP. How do the image encoder and text encoder generate aligned embeddings?
4	Describe the architecture and working mechanism of Stable Diffusion for text-to-image generation.
5	Explain the need for combining vision and language modalities in AI systems.
6	Describe the two main components of the CLIP architecture (image encoder and text encoder).
7	What is contrastive learning, and how is it used to train multimodal models?
8	Explain the basic working principle of text-to-image generation using diffusion models.
9	Describe the concept of a shared embedding space in vision-language models.
10	What is the role of the latent space in Stable Diffusion's image generation process?

L3 Level Questions (Applying / Analyzing)

S.No	L3 - Level Questions
1	How does contrastive learning work in CLIP? Explain how it aligns image and text representations in the same feature space.
2	Explain how CLIP performs image-to-text and text-to-image similarity matching with an example.
3	Compare Stable Diffusion and DALL·E in terms of architecture, training approach, and applications.
4	Design a multimodal AI system using vision-language models for image captioning or visual search. Explain its workflow.
5	Analyze how CLIP's contrastive loss function encourages alignment between matching image-text pairs while pushing apart non-matching pairs.
6	Design a cross-modal retrieval system using CLIP embeddings for searching images using text queries.
7	Compare CLIP-based zero-shot classification with traditional supervised image classification approaches.
8	Evaluate the differences between DALL·E and Stable Diffusion in terms of underlying architecture and generation approach.
9	How would you apply a vision-language model to build an automated image-captioning system for accessibility applications? Explain the workflow.
10	Analyze the challenges involved in aligning visual and textual embeddings in a shared feature space, and how contrastive learning addresses them.

Chapter 7: Retrieval-Augmented Generation

L2 Level Questions (Understanding)

S.No	L2 - Level Questions
1	What is Retrieval-Augmented Generation (RAG)? Explain its importance in improving LLM performance.
2	Explain the architecture of a RAG system with the help of a block diagram.
3	What are embeddings? How do they help in semantic search and retrieval?
4	Explain the role of the retriever component in a RAG pipeline.
5	What is a vector database, and why is it used in RAG systems?
6	Describe the function of the generator component in a Retrieval-Augmented Generation system.
7	Explain the concept of top-k retrieval in the context of RAG.
8	What is the difference between Naïve RAG and Advanced RAG?
9	Explain how LangChain facilitates building a RAG pipeline.
10	Describe the role of the embedding model in converting text into vector representations for RAG.

L3 Level Questions (Applying / Analyzing)

S.No	L3 - Level Questions
1	Explain cosine similarity and its role in vector retrieval systems.
2	Compare traditional keyword-based retrieval with embedding-based retrieval.
3	How does a vector database such as FAISS improve retrieval efficiency? Explain with an example.
4	Explain the process of context retrieval and how it improves response generation in RAG systems.
5	Design a basic RAG pipeline for a domain-specific chatbot application. Explain each component.
6	Differentiate between Naïve RAG, Advanced RAG, and Hybrid RAG.
7	How can RAG be used in real-world applications such as healthcare, legal AI, or academic assistants?
8	Analyze how reranking improves the quality of retrieved context in an advanced RAG pipeline.
9	Design a domain-specific RAG-based question-answering system for a legal-document repository, explaining each architectural component.
10	Evaluate the trade-offs between using FAISS and ChromaDB as vector databases for a large-scale RAG application.